

Hands-on Session on Implementing Cutting Plane Method for TSP

Tarun Rambha

Associate Professor

Centre for infrastructure, Sustainable Transportation and Urban Planning
(CiSTUP)

Robert Bosch Centre for Cyber Physical Systems (RBCCPS)

Indian Institute of Science (IISc) Bangalore

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IIT Delhi

Overview

- ▶ TSP Formulations
- ▶ Blossoms, Combs, and Clique Trees

TSP Formulations

Traveling Salesperson Problem

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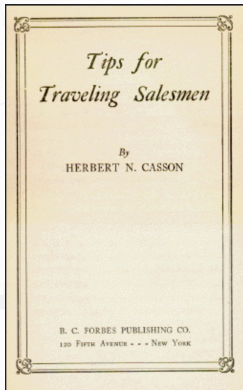
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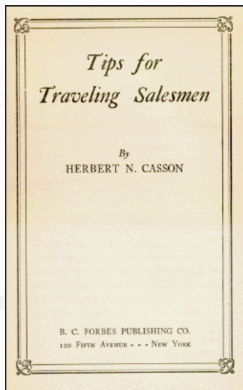
In addition to logistics, TSPs have also been widely used in DNA sequencing and PCB manufacturing.



Traveling Salesperson Problem



Traveling Salesperson Problem



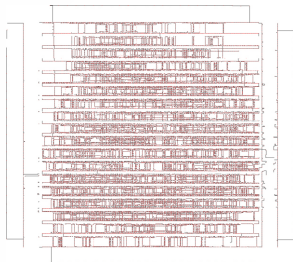
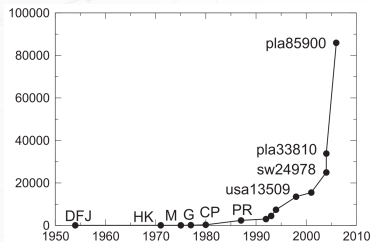
<https://www.math.uwaterloo.ca/tsp/concorde.html>

Traveling Salesperson Problem

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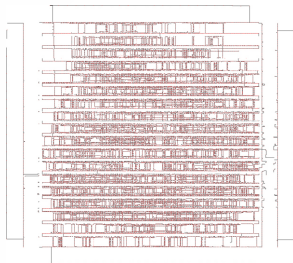
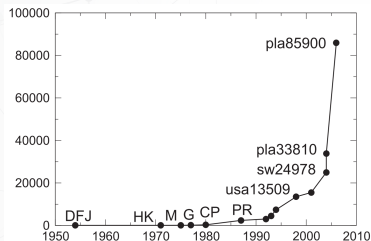
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Several breakthroughs followed in the next few decades. The pla85900 is a chip design instance that was solved optimally in 2006.

Traveling Salesperson Problem

Mona Lisa TSP challenge: 100,000 node instance



From Concorde – July 27, 2012: $GAP=107$. A truncated branch-and-cut search, using an artificial upper bound of 5757092 and the LP from April 18, was terminated after 11.5 CPU years and 20,787 search nodes. The run improved the gap by 4 units.

Traveling Salesperson Problem

Consider an undirected graph $G = (V, E)$ with vertices V and undirected edges E . For simplicity, suppose that G is a complete graph. Every pair of cities connected by an edge $e = \{u, v\}$ has a distance/cost c_e .

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$$\min \sum_{e \in E} c_e x_e$$

$$\text{s.t. } \sum_{u \in e} x_e = 2$$

$$\sum_{u \in S} \sum_{v \in S} x_{\{u, v\}} \leq |S| - 1$$

$$x_e \in \{0, 1\}$$

$$\forall u \in V$$

$$\forall S \subset V, S \neq \emptyset$$

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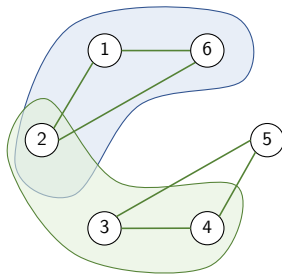
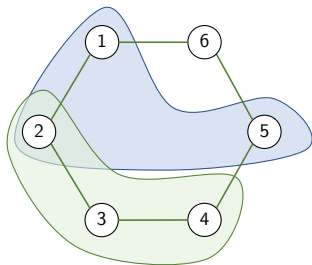
The first constraint ensures that each city is visited exactly once and the second constraint prevents sub-tours.

Traveling Salesperson Problem

What if we did not have sub-tour elimination constraints? When is the constraint strictly $<$ and when is it $=$?

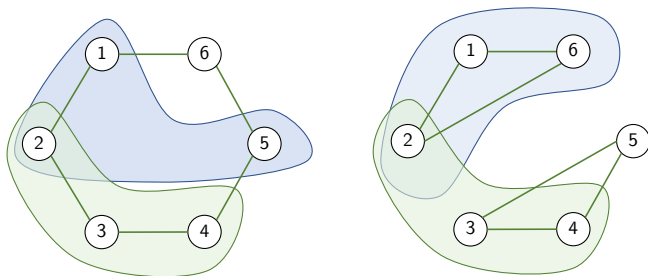
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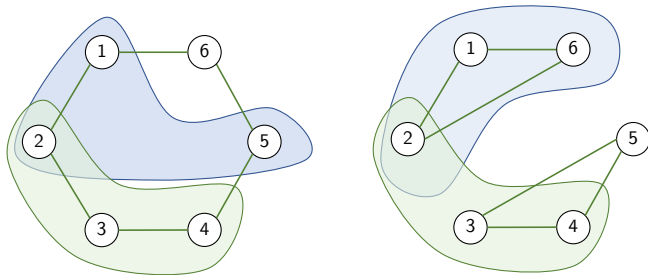


An alternate way of writing the sub-tour elimination constraints (SEC) is

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Apply this version to the above examples? Show that the two SECs are equivalent.

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Additionally, we define $x(E(S))$ and $x(\delta(S))$ as follows

$$x(E(S)) = \sum_{u \in S} \sum_{v \in S} x_{\{u,v\}}$$

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The x s are also called *incidence vectors*. The subgraph with edges for which x values are positive is called the *support graph*.

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The DFJ formulation using E - δ notation can be written as

$$\min \sum_{e \in E} c_e x_e$$

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We can further restrict $3 \leq |S| \leq |V|/2$. (Why?)

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The end result is a graph that is fully connected, but some of the flows can still be non-integral. A new family of cuts are generated using more involved graph theoretical concepts to get to the final optimal solution.

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This motivates the idea of using row generation or lazy cuts (within a branch and cut scheme) for solving the TSP.

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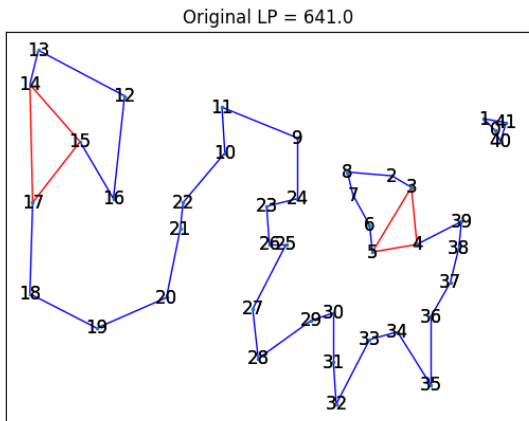
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Suppose we solve the LP relaxation of the TSP problem with just degree constraints. The resulting solution may be fractional or may not be a tour. Hence, the violated constraint can be identified and added iteratively.

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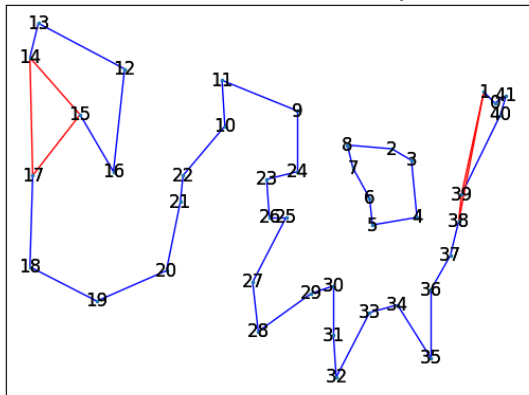
Consider Dantzig's 42-city network. The LP solution lower bound without SECs is 641 (compared to the optimal IP solution of 699) and has fractional variables and is not a tour because of the disconnected components.



Traveling Salesperson Problem

After adding the SEC with $S = \{0, 1, 40, 41\}$, the North-East part of the tour gets connected but a new subtour is created.

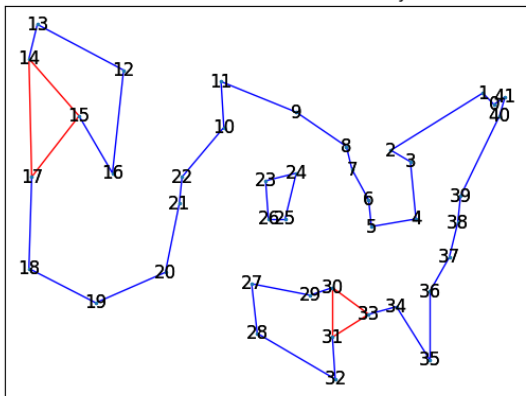
LP solution = 676.0 after connectivity cut 1



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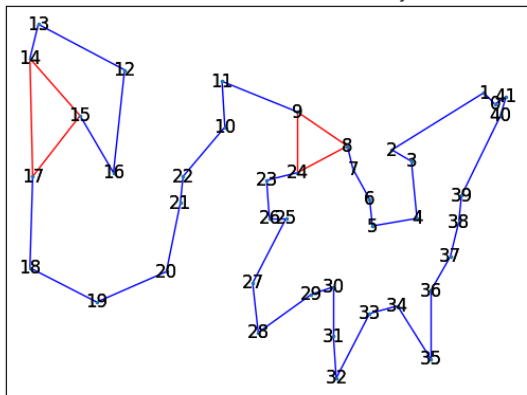
Add a new SEC with $S = \{2, 3, 4, \dots, 8\}$.

LP solution = 681.0 after connectivity cut 2



Add a new SEC with $S = \{23, 24, 25, 26\}$.

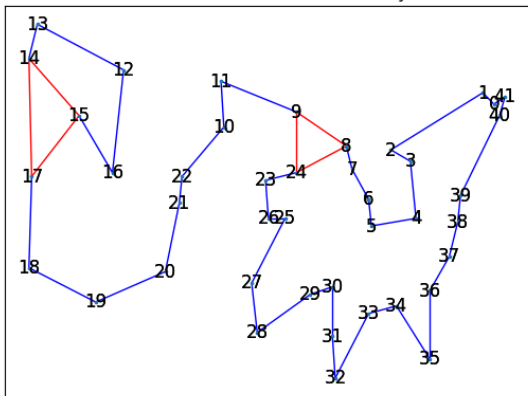
LP solution = 682.5 after connectivity cut 3



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Add a new SEC with $S = \{23, 24, 25, 26\}$.

LP solution = 682.5 after connectivity cut 3



The solution is connected, but is not integral or a tour. Are all SECs satisfied?

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Definition (k -connected)

A graph with at least k vertices is said to be k -connected if it remains connected after removing fewer than k vertices.

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Can you identify articulation points in the previous support graph? To make the graph 2-connected, we can add SECs based on the connected components created after deleting the articulation points.

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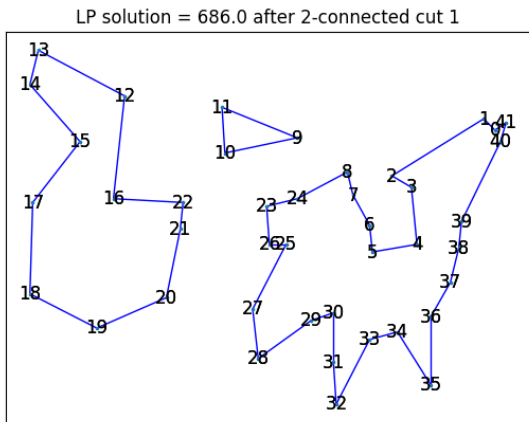
The support graph after adding 2-connected cuts using articulation point 17 has a better LP relaxation but is disconnected. Is this 2-connected?

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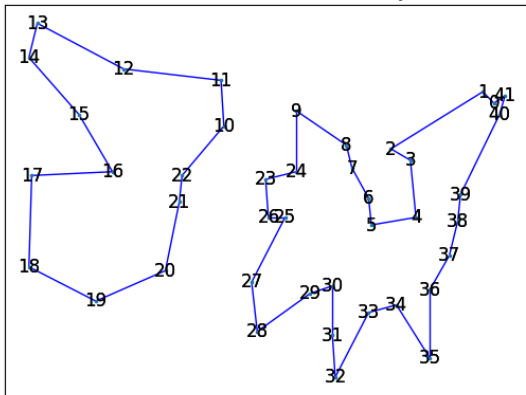
The support graph after adding 2-connected cuts using articulation point 17 has a better LP relaxation but is disconnected. Is this 2-connected? How should we proceed? We can add three connectivity cuts.



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The new solution does not violate the added cuts. Add another connectivity cut.

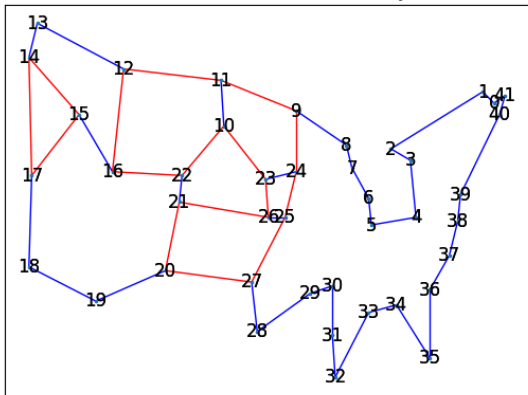
LP solution = 688.0 after connectivity cut 6



Traveling Salesperson Problem

This solution is connected and 2-connected. However, there are many edges with fractional LP solutions.

LP solution = 697.0 after connectivity cut 7

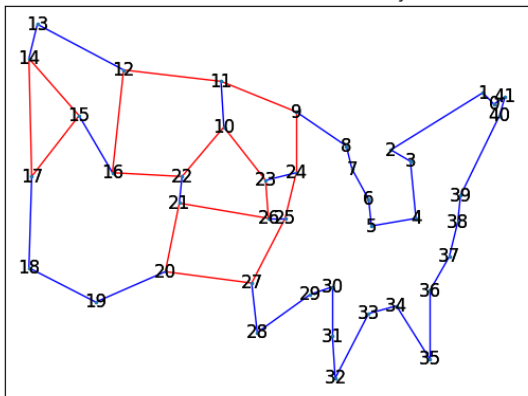


At this stage, one could branch on one of the fractional variables. This may lead to LP relaxations that violate SECs. Hence, connectivity cuts can be added as *lazy cuts*.

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Cuts by solvers are typically grouped into two types:

- ▶ **Lazy Cuts** These constraints are needed by the model but might be excluded since there are a large number of them.
- ▶ **User Cuts** These are not needed for optimality, but they strengthen the formulation by removing some of the LP relaxation region.

Traveling Salesperson Problem

Are there any other types of constraints that can be added to the TSP problem, and can the relaxation solution be improved?

Is there a way to say when one constraint or type of constraint is better than another in a theoretical sense?

Blossoms, Combs, and Clique Trees

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Definition (Dimension)

A polyhedron X is of dimension k , denoted by $\dim(X) = k$, if the maximum number of affinely independent points in X is $k + 1$.

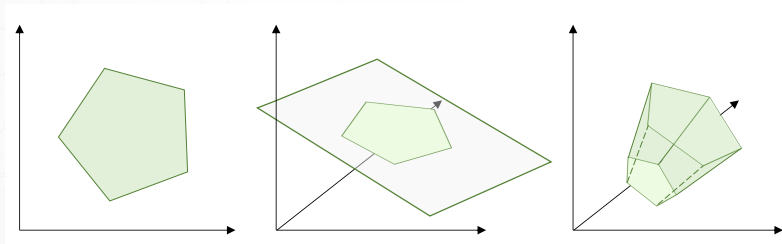
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What are the dimensions of the following polyhedra?



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The dimension of the convex hull of feasible TSP tours is $\binom{n}{2} - n$, where n is the number of vertices.

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Hence, the TSP polytope is not full-dimensional, as expected due to the equality constraints. What is the rank of $(\mathbf{A}^=, \mathbf{b}^=)$?

This makes it harder to prove that certain valid inequalities are facets, and there are infinitely many such valid inequalities.

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Yet, several interesting families of facet-defining valid inequalities are known to exist.

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For example, Grötschel solved a 120-city problem using only 96 facet-defining inequalities.

Theorem (Grötschel and Padberg ('79))

The subtour elimination constraints $x(\delta(S)) \geq 2$ for all $S \subset V$, $3 \leq |S| \leq |V|/2$ are facets of the convex hull of feasible TSP tours.

But what if we can no longer find SECs? Are there other valid inequalities? Are they facets?

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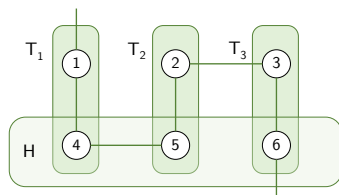
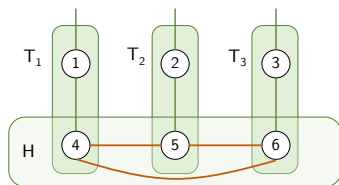
In this example, $T_1 = \{1, 4\}$, $T_2 = \{2, 5\}$, $T_3 = \{3, 6\}$, and $H = \{4, 5, 6\}$. The T sets are also called *teeth* and H is called the *handle*.

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Blossoms, Combs, and Clique Trees

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The RHS is alternatively written as

$$x(E(H)) + \sum_{i=1}^k x(E(T_i)) \leq |H| + \sum_{i=1}^k (|T_i| - 1) - \frac{k+1}{2}$$

Blossoms, Combs, and Clique Trees

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Blossoms, Combs, and Clique Trees

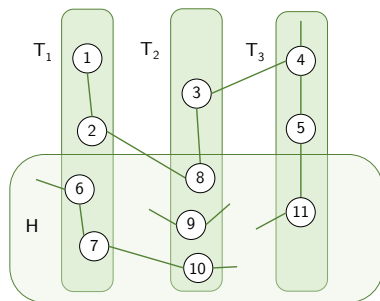
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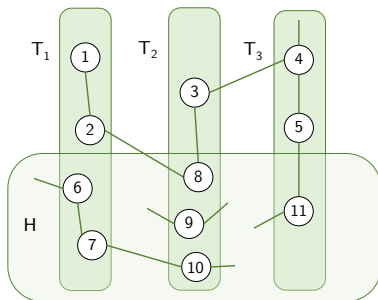
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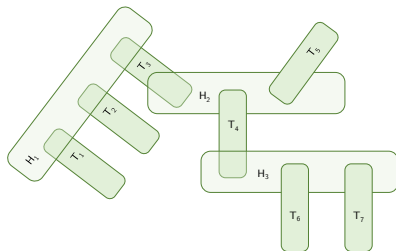
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Thus, every blossom is a comb but not vice versa. The valid inequalities take the same form as described earlier.

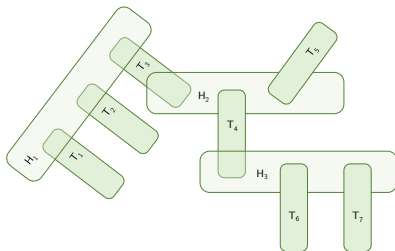
Blossoms, Combs, and Clique Trees

There are extensions of combs called clique trees that can also be used to generate valid inequalities. These have multiple non-intersecting handles and teeth attached to each handle that are non-intersecting as well.



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Proposition (Grötschel & Padberg ('79), Grötschel & Pulleyblank ('86))

Blossom, comb, and clique-tree inequalities are facet-defining for the convex hull of feasible TSP tours.

Unlike some of our earlier examples, proving this result is very challenging since the TSP is not full-dimensional.

Blossoms, Combs, and Clique Trees

Theorem

Every comb inequality is valid for the TSP polytope.

Proof.



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Thus, $x(E(H)) + \sum_{i=1}^k x(E(T_i)) \leq |H| + \sum_{i=1}^k (|T_i| - 1) - \lceil k/2 \rceil$. □

Blossoms, Combs, and Clique Trees

Grötschel and Padberg (1979) then show the following two relatively “simpler” results in a two-part paper.

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To proceed further, they construct four challenging lifting theorems, which lift constraints from lower-dimension to higher-dimensional TSP polytopes.

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Every comb inequality is a facet of the TSP polytope.

Blossoms, Combs, and Clique Trees

The number of combs grows at a much faster rate than the number of sub-tour elimination constraints. A comparison for small-sized problems is shown below.

n	$\nu^S(n)$	$\nu^C(n)$
6	25	60
7	56	2100
8	119	41420
9	246	667800
10	501	8841970
15	16368	1993711339620
20	$0.5 \cdot 10^6$	$1.5 \cdot 10^{18}$
30	$0.5 \cdot 10^9$	$1.5 \cdot 10^{31}$
40	$0.5 \cdot 10^{12}$	$1.5 \cdot 10^{45}$
50	$0.5 \cdot 10^{15}$	10^{60}
59	$0.3 \cdot 10^{18}$	10^{74}
120	$0.6 \cdot 10^{36}$	$2 \cdot 10^{179}$

Blossoms, Combs, and Clique Trees

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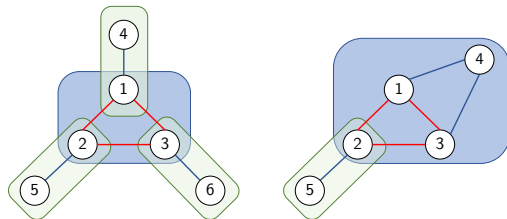
Most codes, however, rely on separation heuristics. For comb inequalities, they typically use network contraction methods (also called safe shrinking) to work with smaller support graphs.

Let us look at the *odd-component heuristic*. Consider a subset of the support graph for which edge weights are fractional. Call this $G_{1/2}$.

Blossoms, Combs, and Clique Trees

Determine the set of connected components in $G_{1/2}$. Let the collection of these nodes be $S_1, S_2, \dots, S_l$.

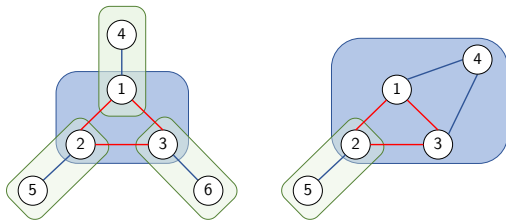
Check if the set $T = \{e : e \in \delta(S_i), x_e^{\text{LP}} = 1\}$ has odd cardinality. The following cases may arise:



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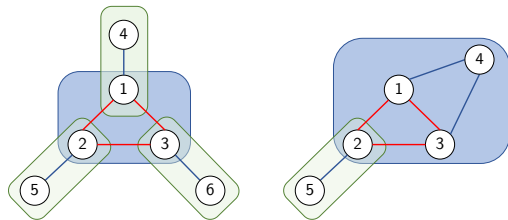


In the first case, edges of T are non-intersecting and hence they naturally form a blossom with $S_i = \{1, 2, 3\}$ as the handle.

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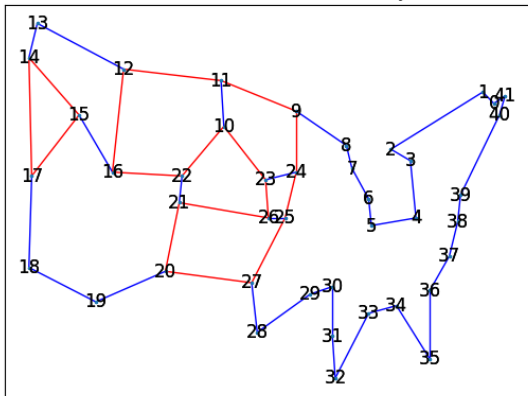
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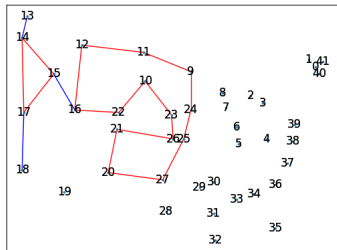
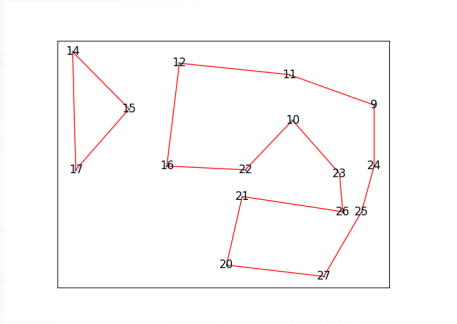
In the second case, edges of T intersect outside S_i , and hence we add the intersecting edge 4 to S_i . $S_i = \{1, 2, 3, 4\}$ provides an SEC.

Blossoms, Combs, and Clique Trees

Can you spot a blossom or comb inequality in the current solution?

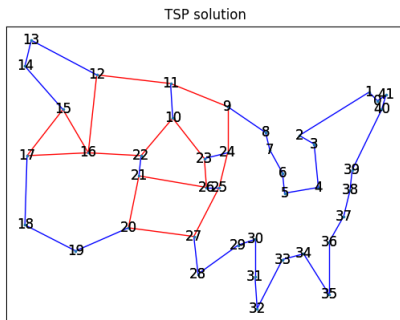
LP solution = 697.0 after connectivity cut 7





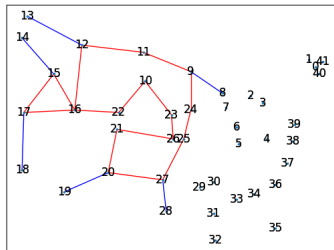
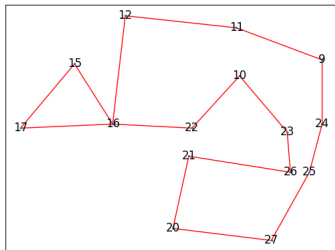
Blossoms, Combs, and Clique Trees

The LP relaxation solution increases to 698 after adding the blossom inequality.



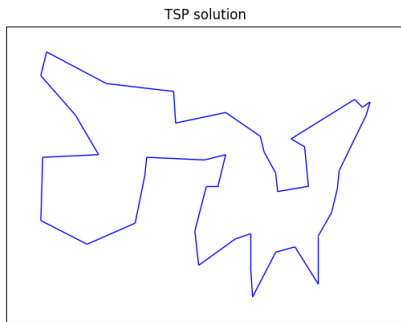
Blossoms, Combs, and Clique Trees

The heuristic cannot be used in this case since there is an even number of teeth.

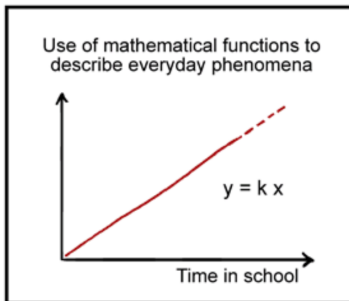


Blossoms, Combs, and Clique Trees

The final optimal tour using branch-and-cut has an objective of 699 and is shown below.



Thank You!



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****footnote:** Thanks to James from U. Wisconsin-Madison

Source: PhD Comics

Questions?